

Algebra II with Trigonometry

Chapter 5.6

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4401272_alg-2trig-h-chp-5-6-note-docx.pdf](#)

Certainly! Here are **10 mental math questions** based on the Law of Cosines and Heron's Formula as reviewed in the PDF. Each question is designed to test understanding of the *core concepts*, not to require lengthy calculation. These are conceptual or use simple numbers, so a student could answer in their head or with quick reasoning.

Law of Cosines and Heron's Formula - Mental Math Questions

1. **True or False:** The Law of Cosines reduces to the Pythagorean Theorem when the included angle is 90° .
2. If the sides of a triangle are 3, 4, and 5 and the angle opposite the side of length 5 is 90° , what type of triangle is it?
3. In triangle ABC, if side $a = 10$, $b = 10$, and the angle between them is 60° , is the triangle equilateral, isosceles, or scalene?
4. Which side of a triangle (a , b , or c) will be the *longest* if the angle opposite to it is the largest?
5. The Law of Cosines calculates an angle when you know all three sides. What information do you need to use the Law of Cosines to find a missing side?
6. If the semiperimeter of a triangle is 10, and the sides are 3, 4, and 5, what is the perimeter?

7. Bearings are measured from which reference directions according to the navigation section? (a) East-West (b) North-South (c) Any direction

8. **Yes or No:** For triangle sides $a = 5$, $b = 5$, $c = 5$, does Heron's formula simplify the calculation of area compared to $A = \frac{1}{2}ab \sin C$?

9. If two sides of a triangle are 7 and 7, and the included angle is 180° , what is the length of the third side?

10. **True or False:** The Law of Cosines can be used for right, obtuse, and acute triangles alike.

Let me know if you want the *answers* or any explanations for these questions!